

CHAPTER 4 Probability Distribution

Consider only

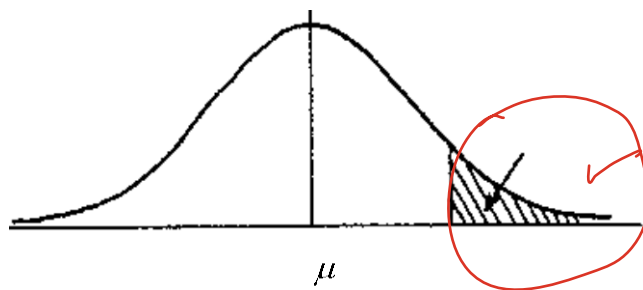
1. Continuous distribution: Normal ✓
2. Discrete distribution: Binomial and Poisson ✓

Normal distribution

A random variable X has a Normal distribution with mean μ and variance σ^2 if and only if its probability function is given by

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right), \quad x \in R, \quad \mu \in R, \quad \sigma > 0$$

and it is denoted by $X \sim N(\mu, \sigma^2)$.

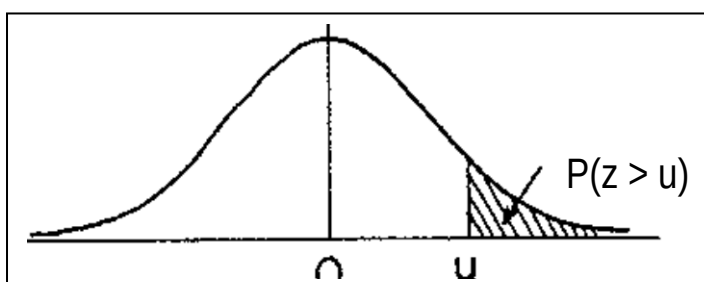


If $X \sim N(\mu, \sigma^2)$, then the random variable $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$. The random variable Z is called the standard Normal distribution with probability function

Handwritten notes:
 Z score
 $0\% = \frac{0}{100} = 0$
 $100\% = \frac{100}{100} = 1$

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right), \quad z \in R$$

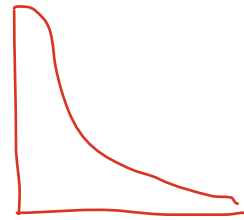
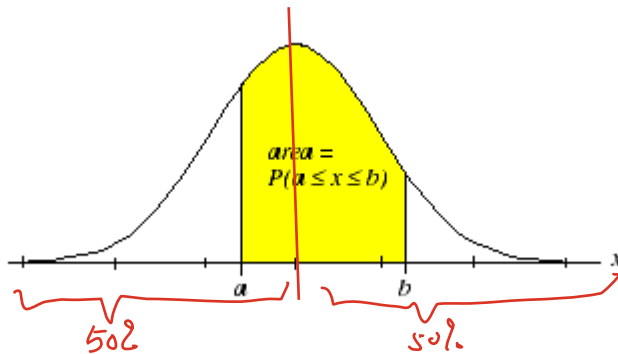
and the curve is symmetry about $z = 0$



Note:

A normal probability distribution, when plotted, gives a bell-shaped curve such that

1. The total area under the curve is 1 $N(0,1)$
2. The curve is symmetric about the mean
3. The two tails of the curve extend indefinitely

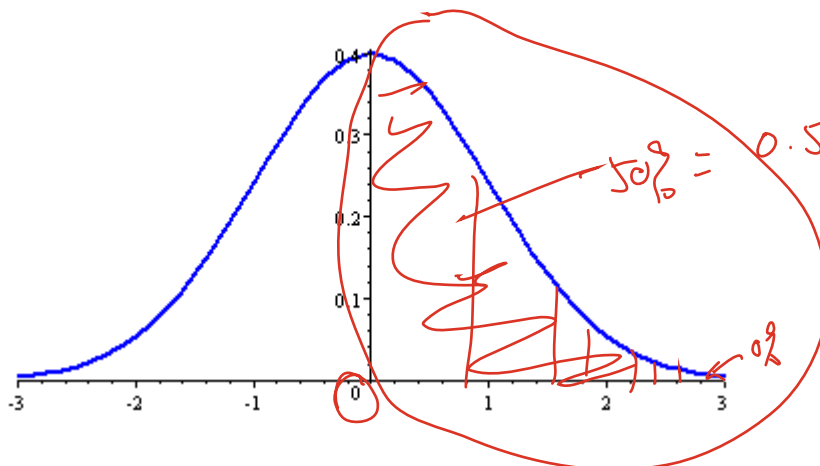


Hence, the area under the curve between two ordinates $X = a$ and $X = b$ where $a < b$, represents the probability that X lies between a and b , denoted by $P(a < X < b)$ $P(a < X < b)$



Note:

In the Normal distribution tables, the partial areas under the Normal curve are tabulated against the standardised variable Z.



Example 1:

Find the following probabilities for the standard Normal curve.

a) $P(Z \geq 1.5)$ $\frac{x-\mu}{\sigma}$

b) $P(Z > -2)$

c) $P(Z < 0.8)$

d) $P(Z < -1.5)$

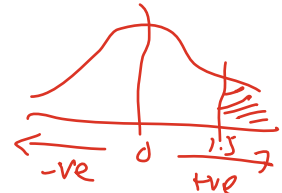
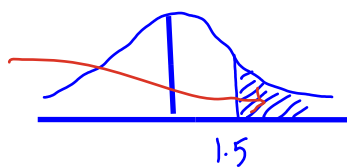
e) $P(-1 < Z < -0.5)$

f) $P(-2 < Z < 1)$

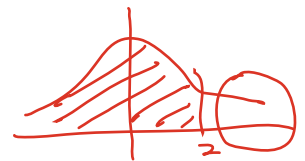
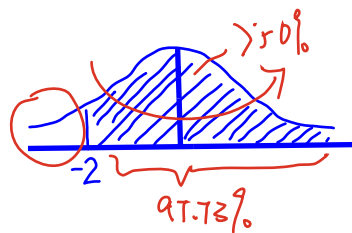
g) $P(1.19 < Z < 2.12)$

Solution:

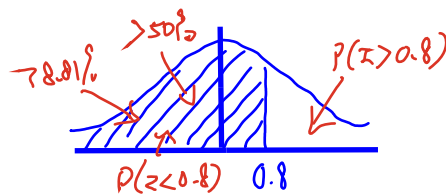
a) $P(Z \geq 1.5) = 0.0668 = 6.68\%$



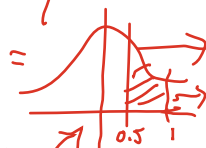
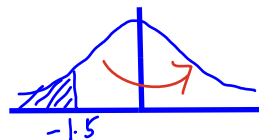
b) $P(Z > -2) = 1 - P(Z < -2)$
 $= 1 - P(Z > 2)$
 $= 1 - 0.02275$
 $= 0.97725$
 $= 0.9773$



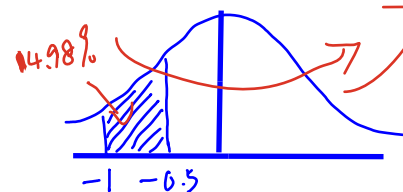
c) $P(Z < 0.8) = 1 - P(Z > 0.8)$
 $= 1 - 0.2119$
 $= 0.7881$



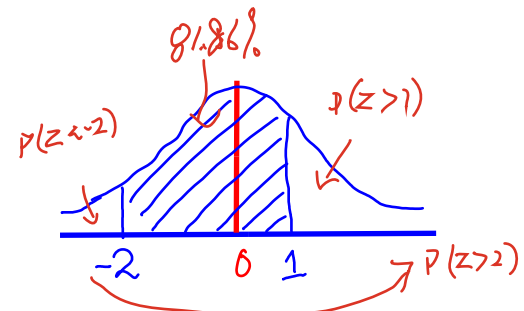
d) $P(Z \leq -1.5) = P(Z > 1.5)$
 $= 0.0668$



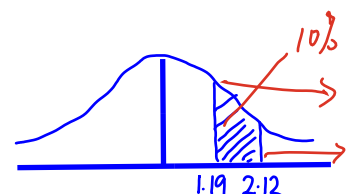
e) $P(-1 < Z < -0.5) = P(0.5 < Z < 1)$
 $= P(Z > 0.5) - P(Z > 1)$
 $= 0.3085 - 0.1587$
 $= 0.1498$



f) $P(-2 < Z < 1) = 1 - P(Z > 1) - P(Z > 2)$
 $= 1 - 0.1587 - 0.02275$
 $= 0.81855$
 $= 0.8186$



g) $P(1.19 < Z < 2.12) = P(Z > 1.19) - P(Z > 2.12)$
 $= 0.1170 - 0.01700$
 $= 0.1$



Example 2:

The mean weight of 200 people is 67kg and the standard deviation is 7kg. Assuming that the weights are Normally distributed, determine how many people have a weight

a) between 60 and 74kg

b) more than 81kg

c) between 53 and 88kg

Solution:

Let $X \equiv$ weight of a person

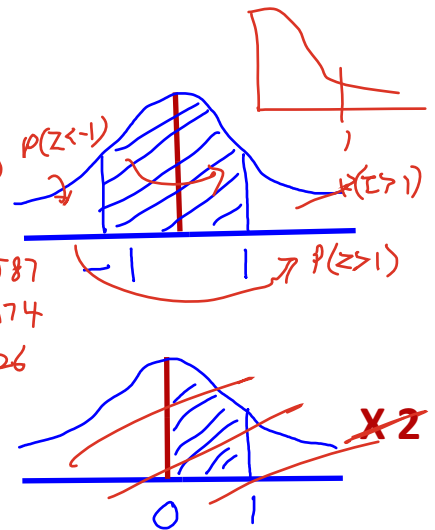
$$\Rightarrow X \sim N(\mu = 67, \sigma^2 = 7^2)$$

$$X \sim N(\mu = 67, \sigma^2 = 7^2)$$

$$\sigma = 7$$

$$-0.5$$

$$\begin{aligned} \text{a) } P(60 < X < 74) &= P\left(\frac{60-67}{7} < Z < \frac{74-67}{7}\right) \\ &= P(-1 < Z < 1) = 1 - P(Z > 1) - P(Z < -1) \\ &= 2 \times P(0 < Z < 1) = 1 - 2P(Z > 1) \\ &= 2 \times [P(Z > 0) - P(Z > 1)] = 1 - 2 \times 0.1587 \\ &= 2 \times [0.5 - 0.1587] = 1 - 0.3174 \\ &= 0.6826 \end{aligned}$$



There are $200 \times 0.6826 = 136.52 \approx 137$ people having a weight between 60 and 74kg.

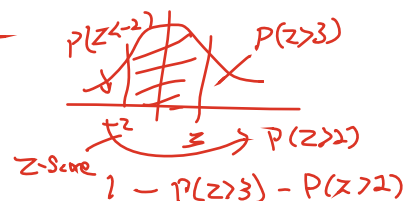
$$\begin{aligned} \text{b) } P(X > 81) &= P\left(Z > \frac{81-67}{7}\right) \\ &= P(Z > 2) \\ &= 0.02275 = 2.275\% \end{aligned}$$



There are $200 \times 0.02275 = 4.55 \approx 5$ people having a weight more than 81kg.

$$\begin{aligned} \text{c) } P(53 < X < 88) &= P\left(\frac{53-67}{7} < Z < \frac{88-67}{7}\right) \\ &= P(-2 < Z < 3) \\ &= 1 - P(Z < -2) - P(Z > 3) \\ &= 1 - P(Z > 2) - P(Z > 3) \\ &= 1 - 0.02275 - 0.00135 \\ &= 0.9759 \end{aligned}$$

$$\frac{x - \mu}{\sigma}$$



There are $200 \times 0.9759 = 195.18 \approx 195$ people having a weight between 53 and 88kg.

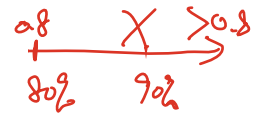
Example 3:

The score on a final examination was normally distributed with mean μ 72 and the standard deviation σ 9. The top 10% of the students are receive A's. What is the minimum score that a student must get in order to receive an A?

Solution:

Let $X \equiv$ score on a final examination

$$\Rightarrow X \sim N(72, 9^2) \quad X \sim N(\mu=72, \sigma^2=81)$$



Let c = minimum score that a student must get in order to receive an A
Given $P(X > c) = 0.10$

$$P(Z > \frac{c-\mu}{\sigma}) \quad P(X > c) = 0.10$$

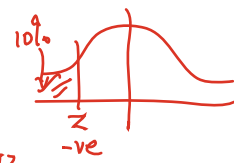
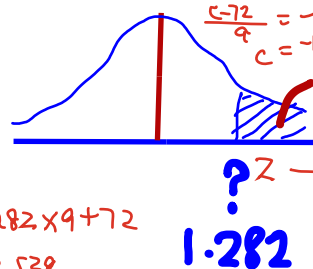
$$P(Z > \frac{c-72}{9}) = 0.10$$

$$\frac{c-72}{9} = 1.282$$

$$c = 83.538$$

$$\approx 84$$

$$c = 1.282 \times 9 + 72 = 83.538 \approx 84 \text{ marks}$$



Minimum score that a student must get in order to receive an A is 84.

Example 4:

$$\frac{X-\mu}{\sigma} = \frac{X-0}{1} = X$$

If $Z \sim N(0, 1)$, find the value of a

a) $P(Z \geq a) = 0.3783$

b) $P(Z \geq a) = 0.7823$

c) $P(Z \leq a) = 0.0793$

d) $P(Z \leq a) = 0.9693$

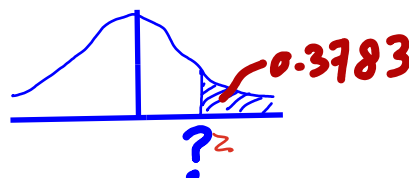
e) $P(|Z| < a) = 0.9$

f) $P(|Z| > a) = 0.0602$

Solution:

a) $P(Z > a) = 0.3783$

$$a = 0.31$$



b) $P(Z > a) = 0.7823$

(flip to right)

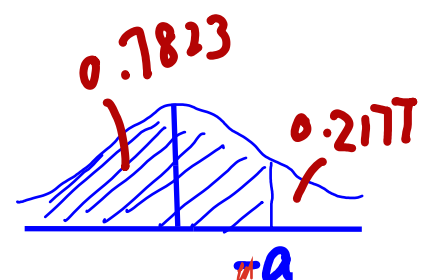
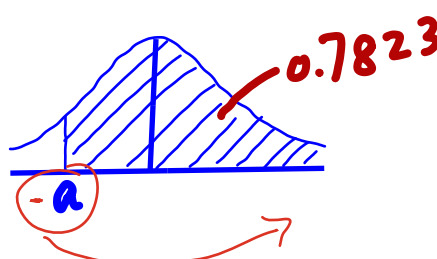
Since the probability is greater than 0.5, a must be negative value

$$P(Z > -a) = 1 - 0.7823$$

$$= 0.2177$$

$$a = 0.78$$

$$a = -0.78$$



$$P(Z > a) = 0.7823$$

$$P(Z < a) = 1 - 0.7823$$

$$P(Z > a) = 0.2177$$

$$\alpha = 0.78$$

c) $P(Z < a) = 0.0793$

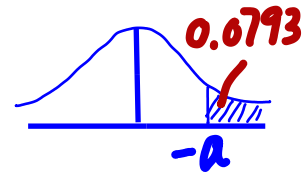
(flip to right)

Since the probability is less than 0.5, a must be negative value

$$P(Z > -a) = 0.0793$$

$$-a = 1.41$$

$$a = -1.41$$

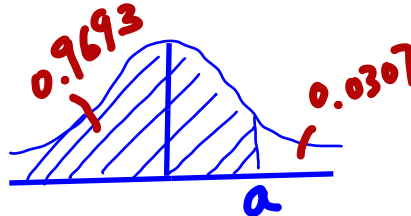


d) $P(Z \leq a) = 0.9693$

$$1 - P(Z > a) = 0.9693$$

$$P(Z > a) = 0.0307$$

$$a = 1.87$$



$$P(Z > a) = \frac{0.1}{2} = 0.05$$

e) $P(|Z| < a) = 0.9$

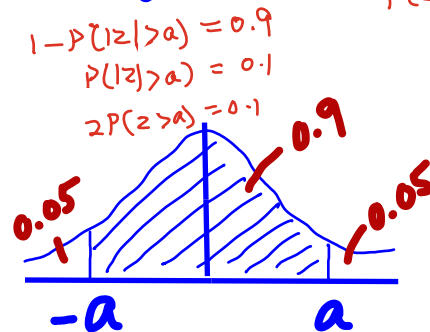
$$P(-a < Z < a) = 0.9$$

$$1 - P(Z < -a) - P(Z > a) = 0.9$$

$$1 - 2P(Z > a) = 0.9$$

$$P(Z > a) = 0.05$$

$$a = 1.645$$



f) $P(|Z| > a) = 0.0602$

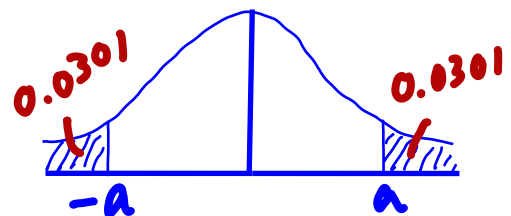
$$P(Z > a) + P(Z < -a) = 0.0602$$

$$P(Z > a) + P(Z > a) = 0.0602$$

$$2P(Z > a) = 0.0602$$

$$P(Z > a) = 0.0301$$

$$a = 1.88$$



Example 5:

If $X \sim N(100, \sigma^2)$ and $P(X < 106) = 0.8849$, find the standard deviation σ .

$$P(X < 106) = 0.8849$$

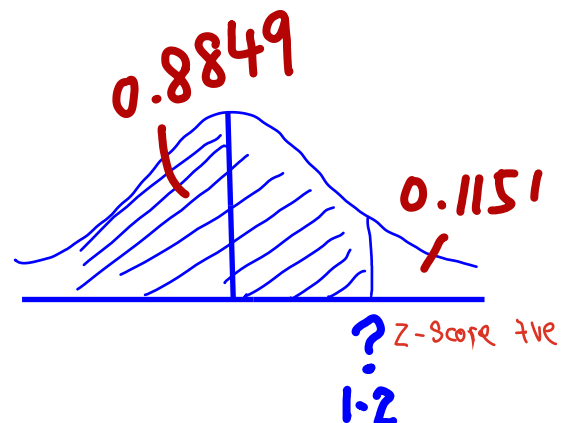
$$P\left(Z \leq \frac{106 - 100}{\sigma}\right) = 0.8849$$

$$P\left(Z < \frac{6}{\sigma}\right) = 0.8849$$

$$P\left(Z > \frac{6}{\sigma}\right) = 1 - 0.8849 = 0.1151$$

$$\frac{6}{\sigma} = 1.2 \rightarrow \sigma = \frac{6}{1.2}$$

$$\therefore \sigma = 5$$



Binomial Distribution

A Binomial experiment must satisfy the following 4 conditions

1. There are n identical trials
→ an experiment is repeated n times under identical conditions
2. Each trial has **only two possible outcomes**
→ success or failure
3. The probabilities of the two outcomes remain constant
→ The probability of success is denoted by p → 0.8
→ The probability of failure is denoted by q ← $(1-p)$ 0.2
→ **$p + q = 1$**
4. The trials are independent
→ Outcome of one trial does not affect the outcome of another trial

Let random variable $X \equiv$ the total number of success in the n trials

→ X is a Binomial distribution with parameter n and p

→ denoted by $X \sim B(n, p)$ and $x = 0, 1, \dots, n$

For a Binomial distribution,

→ mean = np and variance = $np(1-p)$

$$\mu = \frac{\sum x}{N}$$

$$= npq \quad SD = \sqrt{npq}$$

The probability of exactly x successes in n trials is given by

$$P(X = x) = {}^nC_x p^x (1-p)^{n-x}; \quad x = 0, 1, \dots, n; \quad P \in [0, 1]$$

$$\left({}^nC_x \right)^x$$

where

n = total number of trials

p = probability of success

x = number of successes in n trials

nC_x

Example 6:

If the probability of a defective bottle is 0.1, find the mean and standard deviation for the distribution of defective bottles in a total of 400.

Solution

Given $n = 400, p = 0.1$

$$\Rightarrow \text{Mean} = \mu = np = 400(0.1) = 40 \text{ unit}$$

We will expect 40 defective bottles in a sample of 400 bottles

$$\Rightarrow \text{Variance} = \sigma^2 = npq = 400(0.1)(0.9) = 36 \text{ unit}^2$$

$$\Rightarrow \text{Standard deviation} = \sigma = \sqrt{36} = 6 \text{ unit}$$

Example 7:

If 20% of the bottles produced by a machine are defective, determine the probability that, out of 4 bottles chosen at random,

- a) 1, b) 0, c) at most 2 bottles will be defective

Solution: $P(X=x) = {}^nC_x p^x q^{n-x}$ $> \geq < \leq$

Let $X \equiv$ total number of defective out of 4.

$$X \sim B(n=4, p=0.2)$$

$$\begin{aligned} \text{a) } P(X=1) &= {}^4C_1 (0.2)^1 (0.8)^3 \\ &= 0.4096 \end{aligned}$$

$$\begin{aligned} \text{b) } P(X=0) &= {}^4C_0 (0.2)^0 (0.8)^4 \\ &= 0.4096 \end{aligned}$$

$$\begin{aligned} \text{c) } P(X \leq 2) &= P(X=0) + P(X=1) + P(X=2) \\ &= 0.4096 + 0.4096 + {}^4C_2 (0.2)^2 (0.8)^2 \\ &= 0.8192 + 0.1536 \\ &= 0.9728 \end{aligned}$$

$$\begin{aligned} P(X \leq 2) &= 1 - P(X > 2) \\ &= 1 - P(X=3) - P(X=4) \end{aligned}$$

Example 8:

The probability that an entering college student will graduate is 0.4. Determine the probability that out of 5 students

- a) none, b) 1, c) at least 1, d) all will graduate

Solution:

Let $X \equiv$ total number of student that will graduate out of 5

$$X \sim B(n=5, p=0.4)$$

$$\text{a) } P(X=0) = {}^5C_0 (0.4)^0 (0.6)^5 = 0.07776$$

$$\text{b) } P(X=1) = {}^5C_1 (0.4)^1 (0.6)^4 = 0.2592$$

$$\begin{aligned} \text{c) } P(X \geq 1) &= 1 - P(X < 1) \\ &= 1 - P(X=0) \\ &= 1 - 0.07776 \\ &= 0.92224 \end{aligned}$$

$$\text{d) } P(X=5) = {}^5C_5 (0.4)^5 (0.6)^0 = 0.01024$$

Normal approximation to the Binomial distribution

⇒ In a Binomial situation when $np > 5$ and $nq > 5$, the Normal distribution with mean = $\mu = np$ and standard deviation = $\sigma = \sqrt{np(1-p)}$ can be used to approximate the Binomial distribution

⇒ $X \sim B(n, p) \approx X \sim N(\mu, \sigma^2)$ when $np > 5$ and $nq > 5$

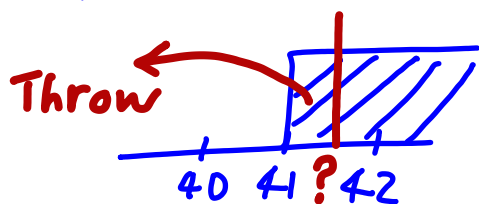
Continuity correction factor ✓

⇒ The addition and/or subtraction of 0.5 from the value(s) of x when the Normal distribution is used as an approximation to the Binomial distribution, where x is the number of successes in n trials

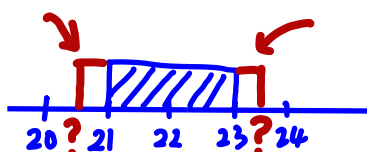
Example 9:

Binomial	Normal
$P(X = 2)$	$P(1.5 < X < 2.5)$
$P(3 \leq X \leq 5)$	$P(2.5 < X < 5.5)$
$P(3 < X \leq 5)$	$P(3.5 < X < 5.5)$
$P(3 \leq X < 5)$	$P(2.5 < X < 4.5)$
$P(3 < X < 5)$	$P(3.5 < X < 4.5)$
$P(X < 4)$	$P(X < 3.5)$
$P(X \leq 4)$	$P(X < 4.5)$
$P(X \geq 4)$	$P(X > 3.5)$
$P(X > 4)$	$P(X > 4.5)$
$P(X \geq 0)$	$P(X > -0.5)$
$P(X > 0)$	$P(X > 0.5)$
$P(X = 0)$	$P(-0.5 < X < 0.5)$

(i) $P(X > 41) \xrightarrow{c.c.} P(X > 41.5)$



(ii) $P(21 \leq X \leq 23) \xrightarrow{c.c.} P(20.5 < X < 23.5)$



Example 10:

Find the probability that 200 tosses of a **bias coin** will result in less than 51 heads in which the probability of getting a head is 0.2 for each toss. $< \leq$

Solution:

Let $X \equiv$ total number of heads in 200 tosses

$$\Rightarrow X \sim B(n=200, p=0.2)$$

$np = 200 \times 0.2 = 40 > 5$
 $npq = 200 \times 0.2 \times 0.8 = 160 > 5$

When $np > 5$, we use normal approximation.

$$\mu = np = 200(0.2) = 40$$

$$\sigma = \sqrt{npq} = \sqrt{200(0.2)(0.8)} = \sqrt{32}$$

$$\Rightarrow X \sim N(40, 32)$$

$$\Rightarrow X \sim N(np=40, npq=40 \times 0.8=32)$$

$$P(X < 51) = P(X < 50.5)$$
$$= P\left(Z < \frac{50.5 - 40}{\sqrt{32}}\right)$$

$$= P(Z < 1.86)$$

$$= 1 - P(Z > 1.86)$$

$$\approx 1 - 0.0314$$

$$\approx 0.9686$$



Example 11:

Find the probability of getting between 3 and 6 heads inclusive in 10 tosses of a **fair coin** by using the Normal approximation to the Binomial distribution. $< \leq > \geq$

Solution:

Let $X \equiv$ total number of heads in 10 tosses

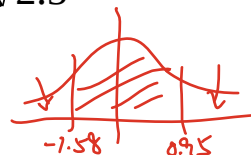
$$\Rightarrow X \sim B(n=10, p=0.5) \Rightarrow X \sim N(np=10 \times 0.5=5, npq=5 \times 0.5=2.5)$$

We can use normal approximation.

$$\mu = np = 10(0.5) = 5$$

$$\sigma = \sqrt{npq} = \sqrt{10(0.5)(0.5)} = \sqrt{2.5}$$

$$\Rightarrow X \sim N(5, 2.5)$$



$$P(3 \leq X \leq 6) = P(2.5 < X < 6.5)$$

$$= P\left(\frac{2.5-5}{\sqrt{2.5}} < Z < \frac{6.5-5}{\sqrt{2.5}}\right)$$

$$= P(-1.58 < Z < 0.95)$$

$$= 1 - P(Z > 0.95) - P(Z > 1.58)$$

$$= 1 - 0.1711 - 0.0571$$

$$= 0.7718$$

Poisson distribution

A discrete random variable having probability function of the form 0,1,2,

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots$$

lambda

where λ can take any positive value, is said to follow the Poisson distribution

Note: If λ is not given, then $\lambda = np$ = mean

If an event is randomly scattered in time (or space) and has mean number of occurrence λ in a given interval of time (or space) and X is the random variable "the number of occurrences in the given interval"

→ X is a Poisson distribution with parameter λ

→ denoted by $X \sim P_o(\lambda)$ and $x = 0, 1, 2, \dots$

For a Poisson distribution with parameter λ ,

→ mean = λ and variance = λ

Example 12:

The mean number of bacteria per ml of a liquid is known to be $\overset{\lambda}{4}$. Assuming that the number of bacteria follows a Poisson distribution, find the probability that, in 1 ml of liquid, there will be

a) no b) 4 c) less than 3 bacteria

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Solution:

Let $X \equiv$ number of bacteria in 1 ml of liquid }
 $\Rightarrow X \sim P_o(\lambda = 4)$

$$a) P(X=0) = \frac{e^{-4} 4^0}{0!} = 0.0183$$

$$b) P(X=4) = \frac{e^{-4} 4^4}{4!} = 0.1954$$

$$\begin{aligned} c) P(X < 3) &= P(X=0) + P(X=1) + P(X=2) \\ &= e^{-4} \left[\frac{4^0}{0!} + \frac{4^1}{1!} + \frac{4^2}{2!} \right] \\ &= e^{-4} [1 + 4 + 8] \\ &= 0.2381 \end{aligned}$$

Example 13:

Based on information obtained in previous example, find the probability that

- a) in 3 ml of liquid, there will be less than 2 bacteria
- b) in 0.5 ml of liquid, there will be more than 2 bacteria

Solution:

- a) In 1 ml of liquid, we expect 4 bacteria

→ In **3 ml** of liquid, we expect $4 \times 3 = 12$ bacteria

Let $Y \equiv$ number of bacteria in **3 ml** of liquid }

$$\Rightarrow Y \sim P_o(\lambda = 12)$$

$$\begin{aligned} P(Y < 2) &= P(Y=0) + P(Y=1) \\ &= e^{-12} \left[\frac{12^0}{0!} + \frac{12^1}{1!} \right] \\ &= e^{-12} [1 + 12] \\ &= 0.00007987 \checkmark \\ &= 0.7987 \times 10^{-4} \end{aligned}$$

- b) In 1 ml of liquid, we expect 4 bacteria

→ In **0.5 ml** of liquid, we expect $4 \times 0.5 = 2$ bacteria

Let $W \equiv$ number of bacteria in **0.5 ml** of liquid

$$\Rightarrow W \sim P_o(\lambda = 2)$$

$$\begin{aligned} P(X > 2) &= P(X=3) + P(X=4) + P(X=5) + P(X=6) + \dots \\ &= 1 - P(X \leq 2) \\ &= 1 - [P(X=0) + P(X=1) + P(X=2)] \\ &= 1 - \left[e^{-2} \left(\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} \right) \right] \\ &= 1 - 0.6767 \\ &= 0.3233 \end{aligned}$$

Poisson approximation to the Binomial distribution

- A Binomial distribution with parameters n and p can be approximated by a Poisson distribution with parameter $\lambda = np$ if n is large and p is small.
- The approximation gets better as $n \rightarrow \infty$ and $p \rightarrow 0$
- $X \sim B(n, p) \approx X \sim P_o(\lambda = np)$ when $n \rightarrow \infty$ and $p \rightarrow 0$

Example 14:

Eggs are packed in boxes of 500. On average, 0.8% of the eggs are found to be broken when the eggs are unpacked. Find the probability that in a box of 500 eggs

- i) less than 2 ii) more than 2 will be broken

Solution:

Let $X \equiv$ number of broken eggs in a box of 500 eggs

$$\Rightarrow X \sim B(n=500, p=0.008) \quad \checkmark \quad np=4 < 5 \quad \times \quad N$$

Since n is large and p is small, we can use Poisson approximation.

$$\lambda = np = 500(0.008) = 4$$

$$\Rightarrow X \sim P_o(4)$$

$$\begin{aligned} \text{i) } P(X < 2) &= P(X=0) + P(X=1) \\ &= e^{-4} \left[\frac{4^0}{0!} + \frac{4^1}{1!} \right] \\ &= 0.0916 \end{aligned}$$

$$\begin{aligned} \text{ii) } P(X > 2) &= 1 - P(X \leq 2) \\ &= 1 - [P(X=0) + P(X=1) + P(X=2)] \\ &= 1 - e^{-4} \left[\frac{4^0}{0!} + \frac{4^1}{1!} + \frac{4^2}{2!} \right] \\ &= 1 - 0.2318 \\ &= 0.7619 \end{aligned}$$

Normal approximation to the Poisson distribution

- If $X \sim P_o(\lambda)$, then mean = $\mu = \lambda$ and variance = $\sigma^2 = \lambda$ $X \sim N(\lambda, \lambda)$
- For large λ , $X \sim N(\lambda, \lambda)$ approximately
- Generally, we **require $\lambda > 20$** for a good approximation
- $X \sim P_o(\lambda) \approx X \sim N(\lambda, \lambda)$ when λ is large

Example 15:

The number of calls received by an office switchboard per hour follows a Poisson distribution with parameter 30. Using the Normal approximation to the Poisson distribution, find the probability that in one hour, there are

a) more than 23 calls b) between 25 and 28 calls inclusive

Solution:

Let $X \equiv$ the number of calls in one hour.

$$\Rightarrow X \sim P_o(30) \quad \lambda > 20$$

× When $\lambda > 20$, we can use Normal approximation

$$\Rightarrow X \sim N(30, 30)$$

$$a) P(X > 23) = P(X > 23.5)$$

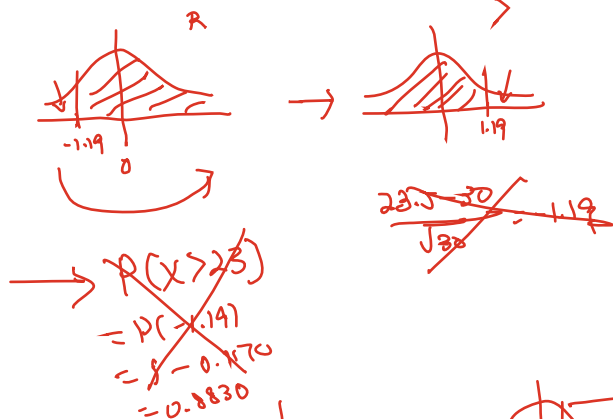
$$= P(Z > \frac{23.5 - 30}{\sqrt{30}})$$

$$= P(Z > -1.19)$$

$$= 1 - P(Z < -1.19)$$

$$= 1 - 0.1170$$

$$= 0.8830$$



$$b) P(25 \leq X \leq 28) = P(24.5 < X < 28.5)$$

$$= P\left(\frac{24.5 - 30}{\sqrt{30}} < Z < \frac{28.5 - 30}{\sqrt{30}}\right)$$

$$= P(-1 < Z < -0.27)$$

$$= P(Z > 0.27) - P(Z > 1)$$

$$= 0.3936 - 0.1587$$

$$= 0.2349$$

